

SinhalaJournal-LLM

A Style-Adaptive Large Language Model Framework for Sinhala Journalism

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering
Project ID: R26-SE-037

Member 1 – Sinhala Grammar Checker
Member 2 – Sinhala Headline & Image Generator
Member 3 – Sinhala News Summarizer
Member 4 – Sinhala Article Style Rewriter

Research Team



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SinhalaJournal-LLM

Product: **SinAi**
Sinhala AI Journalist

Sinhala journalists still have limited AI support for daily newsroom writing.



Grammar correction



Headline writing



News summarization



Style rewriting

1,031,456

Sinhala news articles collected



665,887

Articles retained after cleaning

4

Journalism tasks in one system

One Sinhala AI system for common newsroom writing tasks

Research Gap → Four Objectives

RESEARCH GAP Existing Sinhala NLP tools do not fully support controlled newsroom writing across these four tasks in one research framework.



Grammar

Existing Sinhala grammar correction is mainly rule based or narrow domain, with limited context aware LLM correction for journalism.



Headline

Sinhala headline generation exists, but controlled newsroom generation with reliable length and output constraints remains limited.



Summary

Existing Sinhala summarization is mainly extractive, while controlled abstractive news summarization still needs stronger evaluation.

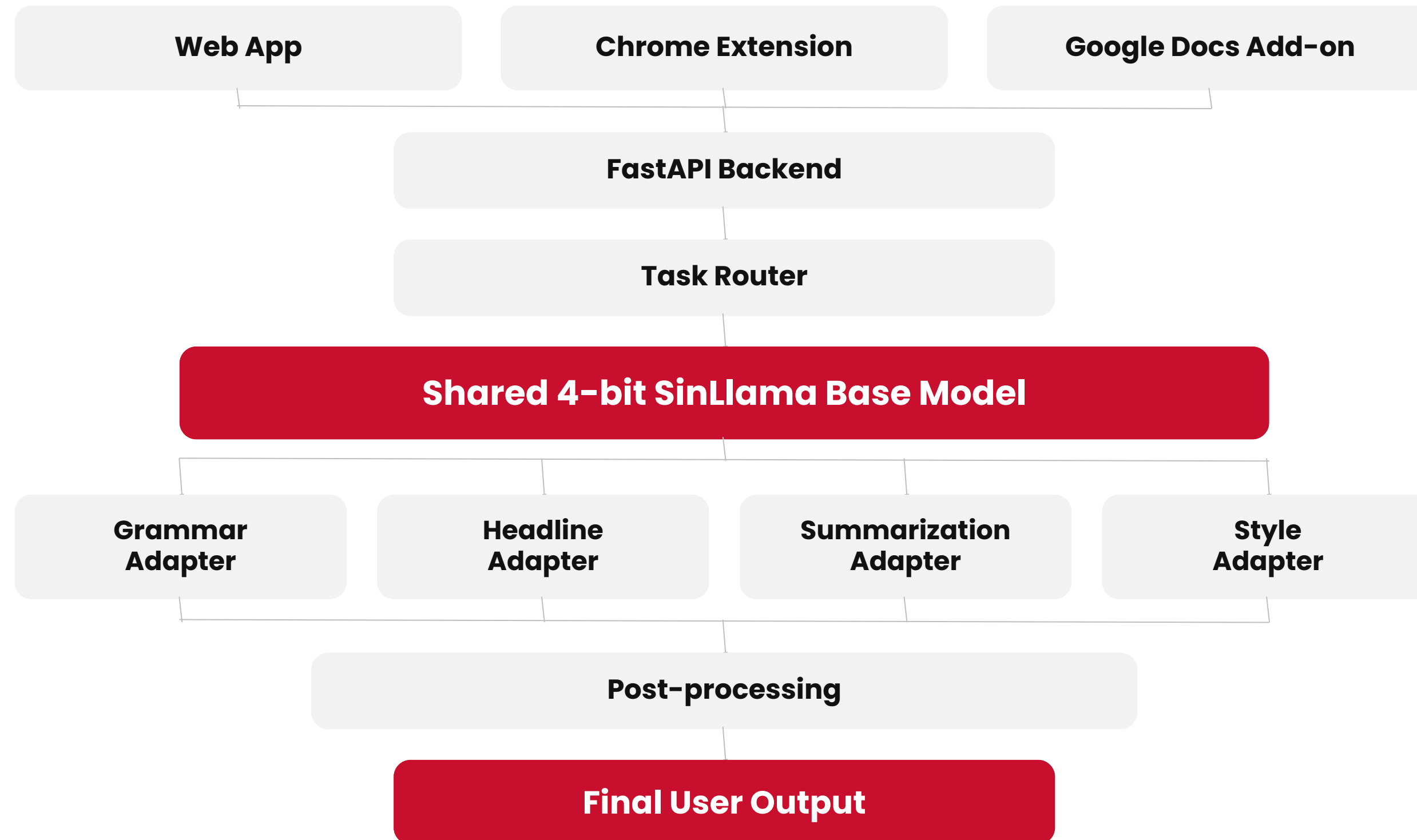


Style

Sinhala has limited datasets and validated systems for rewriting the same news content across multiple journalistic styles.

Shared SinLlama Foundation + Task-Specific LoRA Adapters

One Foundation — Four Specialized Adapters



Why LoRA?

- Shared base model
- Less training cost
- Separate task specialization
- Easier adapter updates

One shared base model. Four small task adapters. One backend.

From Research Models to One Working Product

4 / 4

Components integrated



User Interfaces

- Web application
- Chrome extension
- Google Docs add-on



Engineering

- FastAPI backend
- Shared model server
- Authentication and roles
- Modular services



Quality

- Integration testing
- Regression testing
- Task-specific validation
- Error handling

All four research components are integrated into one working product.

Evaluation Evidence



Grammar

286 frozen evaluation inputs

63 / 143

SinLlama (final)

12 / 143

ByT5-small

0 / 143

mT5-small

Faulty inputs corrected exactly. Much stronger than the evaluated small multilingual baselines.



Headline

Unwanted media artifacts

11.2%

Before



1.1%

After

Target cleaning greatly reduced media artifacts. Controlled headline length stays a key evaluation target.



Summarization

Controlled length targets

Short

Medium

Long

v06 and v07 — no clear overall winner

Evaluation design changed the interpretation of model performance.



Style Rewriting

Meaning Preservation



Style Transformation



Human Evaluation — ONGOING

Final human-rated results are not reported yet.

Better data quality and evaluation design improved the reliability of our findings.

From Research to SinAI



Target Users

- Newsrooms
- Digital publishers
- Journalism students
- Editors



User Value

- Sinhala-first
- Journalism focused
- Four tools in one workflow
- Multiple user interfaces



Possible Commercial Model

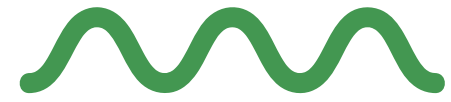
- B2B SaaS
- API access
- Institutional plans

LIVE DEMO

WEBAPP:chat.sin-ai.app

PORTFOLIO:sin-ai.app





**Thank
You**

